MASTER RESEARCH DOSSIER

A consolidated record of nine research programs, formal proofs, simulations, exact certificates, preregistrations, CI verifications, and open frontiers

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Fresh synthesis date: 24 October 2026 (superseding 1 October & 9 October 2026 editions)

Purpose & Epistemic Governance

A reader-facing master document that consolidates the complete public and frozen research record through 24 October 2026 across nine research and verification tracks. It is not a replacement for the primary GitHub repositories, frozen OSF and Zenodo deposits, Lean 4 proof packages, Arb interval certificates, or machine-readable SHA-256 manifests. Where claims differ in strength, the narrower evidence class controls.

Core Research Axiom: *Build. Attack. Falsify. Preserve.* Freeze inputs prior to execution, separate finite computer-assisted certificates from continuum analytical claims, preserve prospective failures unchanged in the ledger, and bind every public statement to reproducible code and cryptographic digests.

Umbrella repository: github.com/reggaesharkk/Reggae-shark-universe- | **Author profile:** github.com/reggaesharkk

This portfolio does not claim that the nine programs form one unified law of nature.

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1. Executive research map

As of 24 October 2026, the portfolio comprises nine distinct research and verification programs. Their unifying contribution is methodological rather than a single physical postulate: freeze inputs before testing, enforce explicit finite claim boundaries, preserve failed prospective gates alongside positive theorems, separate evidence classes, and anchor public releases to deterministic replay suites and cryptographic manifests.

#	Program	Core object	Current state (24 Oct 2026)	Evidence class
01	**Self-Referential Processing + Design Lab**	Preregistered 260-item behavioral protocol (v8.6.3) + 4,000-fit GLMM simulation architecture validation	Phase 7 complete (M1 selected); v8.6.3 live collection paused pending §6.4 API snapshot binding	Preregistered / Simulated
02	**Ψ Self-Modeling Benchmark**	Double-blinded self-vs-peer 2AFC diagnostic benchmark (p = 0.5) & Ψ ∈ [0, 1] metric	v4 pre-confirmatory diagnostic archived; additive OSF manifest hash audit complete	Diagnostic / Preregistered
03	**Information-Theoretic Physics (Appendix A v1.5)**	3D (γ, H_{obs}, $\dot{I}$) soft-wall ODE + 6-face Lean 4 box + exact $\mathbf{Q}(i\omega_c)$ ℓ_1 & 200-digit `mp.iv` audit	v1.5 Lake & CI Run #39587627124 verified; conditional HRT Gate 0 dictionary; general Gate 0 open	Proved / Certified / Reproduced / Open

#	Program	Core object	Current state (24 Oct 2026)	Evidence class
04	**LRSC Odd-Ring Spectral Rank**	Odd-ring single-step rank(V) = p theorem + exact finite M = 99 subset certificates	v1.2 theorem + v1.2.2 errata; M = 99 $K_{0.001}$ = 11 (θ = 0.08) & $K_{0.001}$ = 14 (θ = 0.07) certified	Proved / Certified (Finite)
05	**Navier–Stokes Bridge Audit**	Periodic 3D Fourier–Galerkin audits, prospective K36 gates, 128/192-bit Arb $N_{11} \to N_{18}$ chain	N = 11..18 same-datum chain certified; v0.28 adjoint sign fixed; PR #155 M15 remainder audited; continuum open	Certified (Finite) / Falsified / Open
06	**IntentSeal + VRK v0.3.1**	Fail-closed tool-call mediation prototype + proof-carrying claim/evidence verification kernel	IntentSeal v0.1.1–v0.5.0 frozen; VRK v0.3.1 final (30/30 tests, 33/33 adversarial rejections)	Software Verification Infrastructure
07	**Riemann Bandwidth-One Verification (WP84)**	Discrete Loewner $12/(\pi^2 R)$ trace bound, exact C_5 151-vertex polyhedral sign, conditional zero models	Public v0.1.0 (38 files, 21/21 CI jobs, Zenodo 23244818); A1294 replay; >79% & RH open	Proved / Certified / Conditional / Open
08	**Theory of Bounded Existence (Appendix B v1.0)**	4D (γ, H_{obs}, $\dot{I}$, Ψ) soft-wall ODE + 17-theorem Lean 4 box/floors + 20D variational monodromy	v1.0 released (Zenodo `23608244`, CI Run #6 green, commit `c690801`); P_0 = 0 Ψ-memory bistability proved	Proved / Certified / Reproduced
09	**STRENGTH H4-BRIDGE-1 Audit & 7-Problem Board**	Analytical BPTT & $D-2$ causal SNR audit of 100 RNN16/GRU9 networks + 7-problem falsification map	`H_BPTT_BUG` refuted; Adam v_k clamping & lag bug isolated; prospective LOSO H4 protocol open	Diagnostic Audit / Falsified / Open

Most important global boundary (The Academic Firewall)

No artifact in this portfolio claims a solution to the Riemann Hypothesis, the 3D continuum Navier–Stokes Millennium Problem, P ≠ NP, the Yang–Mills mass gap, Quantum Gravity, Dark Matter, Dark Energy, or machine consciousness/qualia. The strongest results in LRSC, Navier–Stokes, and Riemann WP84 are finite or operator-specific and explicitly scoped. The dynamical theorems in Programs 03 and 08 are internal mathematical properties of dimensionless phenomenological ODE families with no physical energy budget or clinical EEG equivalence.

2. Evidence classes and research discipline

Class	Meaning in this portfolio	Representative ground-truth artifacts
PROVED / ANALYTIC	Mathematical proposition derived under stated assumptions via closed-form algebra or Lean 4 proof kernel.	LRSC rank(V) = p; Loewner $12/(\pi^2 R)$ bound; Programs 03 & 08 Lean 4 pointwise box/floor theorems; $de_c/dP_0 > 0$.
CERTIFIED / COMPUTER-ASSISTED	Finite claim closed via exact rational arithmetic (`fractions.Fraction`), outward interval enclosures (`Arb` / `mp.iv`), or exhaustive enumeration.	LRSC M = 99 $K_{0.001}$ = 11 (13.43B subsets); Navier $N_{11} \to N_{18}$ Arb chain; Riemann C_5 151-vertex & A1294 dyadic replay; Physics v1.5 200-digit `mp.iv` $\ell_1 > 0$.

Class	Meaning in this portfolio	Representative ground-truth artifacts
REPRODUCED / CI-VERIFIED	Pinned multi-solver or multi-platform workflow passing deterministic replay and hash verification.	Riemann v0.1.0 (21/21 CI jobs); Physics v1.5 (CI Run #39587627124); Bounded Existence v1.0 (CI Run #6, commit `c690801`, digest `80e0724e...`).
SIMULATED	Operating characteristics evaluated under a synthetic data-generating process (DGP) with fixed seeds.	Design Lab Phase 7 Monte Carlo GLMM selection (4,000 fits across 500 seeds).
PREREGISTERED	Hypotheses, estimands, seeds, gates, and analysis scripts locked and hashed before confirmatory data.	Self-Referential Processing v8.6.3 protocol (OSF `37H2Y`).
DIAGNOSTIC	Controlled exploratory benchmark or mechanistic code audit not promoted to confirmatory science.	Ψ v4 2AFC benchmark; STRENGTH H4-BRIDGE-1 BPTT & gradient-interference audit.
CONDITIONAL FRONTIER	Exact mathematical implication proved assuming an unproved upstream asymptotic or physical bridge.	Riemann $R = 400$ (68.8203%) and Christoffel 11061/13891 (79.6271%) consumers; Physics v1.5 Proposition 1 HRT dictionary.
FALSIFIED / FAILED / UNEVALUABLE	A frozen prospective gate or proposed shortcut failed verification and is preserved without retuning.	Navier N17 mechanism gate; BGT critical-space gate; Riemann $p_s \le F_{61}$ & 4th-cumulant Gram shortcuts; H4-BRIDGE-1 high-noise J75 gate.
OPEN	A required analytical bound, continuum bridge, or operational deployment condition is not yet closed.	Navier recursive backreaction drift; Riemann signed high-product & Weil transfer; non-geometric Gate 0; H4 LOSO outcome test.

- **Immutable baselines & additive provenance:** Frozen artifacts remain byte-identical; errata, hash audits, and scope clarifications are published additively rather than overwriting historical records.
- **No silent promotion across the epistemic gap:** Numerical persistence across finite Galerkin cutoffs ($N = 11..18$) or high-precision sweeps is never promoted to a continuum PDE or analytic theorem without a separate uniform bridge proof.
- **Preservation of negative evidence:** Failed prospective gates (such as the N17 directional-split mechanism gate or the H4-BRIDGE-1 high-noise J75 gate) remain permanently in the repository record.
- **Source-free & multi-solver verification:** Decisive finite certificates are verified via standalone standard-library runners or cross-checked across independent algorithms (e.g., Python vs. C++ bitmask traversal; `LSODA` vs. `DOP853`; `SymPy` vs. `mpmath.iv`).

3. Program 01: Self-Referential Processing + Design Lab

3.1 Frozen confirmatory protocol v8.6.3 & §6.4 snapshot audit

Research question: Whether large language model performance changes under self-referential versus peer-referential framing in a controlled item-level two-alternative design, after controlling for stance distance, context length, presentation order, scale, and refusal exclusion.

- **Item pool & domains:** 260 items across five domains (AI governance; bioethics and medical rights; data privacy versus public safety; environmental policy and resource allocation; economic trade-offs and automation), split across stance self-consistency and four-layer scrubbed authorship attribution.
- **Portfolio design:** 20-model portfolio (12 open-weight and 8 closed-weight positions) with deterministic round-robin peer assignment.
- **Gate 5 confirmatory estimand:** Item-level linear probability regression (`Perf ~ X_j + ContextLen + Is Self + Order + StanceDist_C`) with Cameron–Gelbach–Miller (CGM) two-way cluster-robust standard errors clustered by `model_id` and `item_id` (SESOI b_3 = 0.10, α = 0.05, 5% refusal exclusion threshold).
- **Calibration & operational status:** Recovery-harness calibration recorded a **6.3%** null rejection rate across 1,000 seeds, inside the prespecified [3%, 9%] band. A 15 September 2026 vendor pre-flight audit (`api_ping_check.py`) found 3 of 8 closed API snapshots live and 5 retired, requiring a §6.4 successor amendment before live collection. Live confirmatory collection remains fail-closed (paused/not started).

Frozen foundation DOI: https://doi.org/10.17605/OSF.IO/37H2Y | **Consolidated Design Lab DOI:** https://doi.org/10.17605/OSF.IO/NV7DZ
Repository: https://github.com/reggaesharkk/self-referential-processing-designlab

3.2 Phase 7 inferential validation

Phase 7 is a separate prospective simulation validation study testing candidate mixed-effects architectures on paired synthetic datasets. It does not amend v8.6.3 retroactively and does not establish an empirical self-effect.

Design parameter	Frozen Phase 7 value	Cell / Architecture	Reject / Power	95% Coverage & Gate
Generated dataset scale	100 models × 400 items = 40,000 obs	Cell A (β_{Self} = 0, ρ = 0) — M1	6.6% null reject	93.4% — **PASS**
Production replicates	500 deterministic seeds / cell	Cell A (β_{Self} = 0, ρ = 0) — M2	7.2% null reject	92.8% — **FAIL**
DGP cells	4 ($\beta_{Self} \in \{0, 0.10\}$ × ρ ∈ {0, 0.30})	Cell B (β_{Self} = 0, ρ = 0.30) — M1	6.4% null reject	93.6% — **PASS**
Candidate architectures	M1 (correlated RE) vs. M2 (uncorrelated RE)	Cell B (β_{Self} = 0, ρ = 0.30) — M2	7.4% null reject	92.6% — **FAIL**
Total GLMM fits	4,000 paired fits (`lme4` recovery verified)	Cell C (β_{Self} = 0.10, ρ = 0) — M1	99.6% power	94.2% — **Coverage pass**
Locked decision rule	Exact-binomial Type-I & 95% coverage screen	Cell D (β_{Self} = 0.10, ρ = 0.30) — M1	99.6% power	94.2% — **Coverage pass**

Phase 7 conclusion: Under the locked screening rule, M1 passes null Type-I calibration and coverage in both covariance-generating conditions while M2 fails. M1 is therefore the candidate architecture for a future preregistered design, conditional on the tested Phase 7 DGP.

4. Program 02: Ψ Self-Modeling Organization & Diagnostic Benchmark

The Ψ Self-Modeling program comprises an operationalized mathematical definition of self-modeling organization ($\Psi \in [0, 1]$) and a distinct pre-confirmatory diagnostic behavioral benchmark (v4) using a deterministic self-versus-peer two-alternative forced-choice (2AFC) structure with chance baseline $p = 0.5$.

- **Purpose & architecture:** Isolates narrow self-versus-peer processing contrasts under deterministic seeding, anonymized presentation, style controls, and negative controls, while defining the dimensionless state-estimation readout Ψ used downstream in Program 08.
- **Provenance discipline:** An integrity audit identified hash mismatches for `README.md` and manuscript PDF files in the initial bundle; the original OSF manifest was preserved intact as historical evidence alongside an additive correction ledger.
- **Claim boundary:** Benchmark performance and Ψ readout values quantify functional state-estimation contrasts only; they are not evidence of machine consciousness, subjective selfhood, or sentience.

OSF DOIs: 10.17605/OSF.IO/N3DQU | 10.17605/OSF.IO/8CFQW | 10.17605/OSF.IO/CAE2H
Internet Archive: archive.org/details/psi-self-modeling-organization | **Repository:** github.com/reggaesharkk/psi-self-modeling-benchmark

5. Program 03: Information-Theoretic Physics & Appendix A v1.5 Soft-Wall Completion

This program strictly separates a specified 3D phenomenological ODE system (γ, H_{obs}, $\dot{I}$) from broader physical or cosmological interpretation. In October 2026, following an 11 October scope audit of the unregularized v1.4 system, the dynamical sector was upgraded to **Appendix A v1.5** (21 October 2026), closing post-Hopf finite-time escape via smooth hyperbolic-tangent soft walls verified in Lean 4 and 200-digit interval arithmetic.

5.1 Lineage from Physics v1.0 & Appendix A v1.4 to the 11 October 2026 scope audit

- **Parent Physics v1.0 & Appendix A v1.4/v1.4.1:** Formulated the unregularized 3D ODE family with e-driven Hopf threshold $e_c = (\alpha + g)(\alpha + \mu)(g + \mu) / [\beta\delta(\alpha + \mu + g)]$ and $\omega_c^2 = \alpha\mu g / (\alpha + \mu + g)$, supported by a 122,378-point numerical sweep and a 93-numerator / 28-denominator positive monomial expansion.
- **Separate three-state (R, Ψ, Γ) cosmology sandbox audit:** Established unique equilibrium and local asymptotic stability under positive parameters, while exposing three internal falsifiers: (i) the normalized Π score can yield $\Pi = 1$ for a contraction $C(U) = qU$ ($0 < q < 1$) despite $C(U) \neq U$; (ii) fresh Gaussian noise makes a single Π^* sample stochastic rather than a theorem; and (iii) a plotting-axis bug plotted a state coordinate instead of time.
- **11 October 2026 scope clarification & post-Hopf escape:** An internal audit noted that in the unregularized v1.0–v1.4 ODE, trajectories with $e > e_c$ escape the positive orthant ($\gamma(t) < 0$ at $t \approx 55$ on the v1.1 benchmark) and required an explicit, machine-checked normal-form derivation of $\ell_1 = \mathrm{Re}(G_{21})/(2\omega_c)$.

5.2 Appendix A v1.5 soft-wall dynamical completion (21 October 2026 release)

Appendix A v1.5 introduces smooth (C^∞) hyperbolic-tangent gates $\Psi_{\varepsilon_H}(H_{obs}) = \tanh(H_{obs}/\varepsilon_H)$ on the irreversible erasure sink and $\Phi_{\varepsilon_I}(\dot{I}) = \tanh(\dot{I}/\varepsilon_I)$ on the derivative-feedback channel:

- **Lean 4 6-Face Compact Trapping Box (`BoundaryVerification.lean`):** Formalizes the vector field over Mathlib's **R** (`leanprover/lean4:v4.36.0-rc1`, Mathlib `10156080d0e7`, verified via `lake build` in GitHub Actions Run #39587627124, commit `4211c6f`) and proves all 9 pointwise face/ceiling theorems and `compact_box_nagumo_certificate` on $B(e) = [0, \gamma_{max}] \times [0, H_{max}] \times [0, \dot{I}_{max}(e)]$ ($\dot{I}_{max}(e) = (eM_H + fP_{input})/g$, $M_H = c\gamma_{max}H_{max} + \dot{I}_{erase}$) with `0 sorry` and `0 nlinarith`.
- **Exact Interior Equilibrium & SymPy Tensor Identity:** Because $\dot{H}_{obs}^* = 0$ at any steady state, $\partial_i[e \tanh(\dot{I}/\varepsilon_I)\, \dot{H}_{obs}]_* = 0$ vanishes identically, preserving $\dot{I}^* = fP_{input}/g$ and $\gamma^* = K\gamma_{max}/(b\gamma_{max} + K)$ exactly. Independent `SymPy` symbolic differentiation verifies $f_1 = f_2 = f_3 = 0$ and zero symbolic residual across $J_{v1.5}$, $B_{v1.5}$, $C_{1,v1.5} = 0$, $C_{2,v1.5}$, $C_{3,v1.5}$.
- **Exact $\mathbf{Q}(i\omega_c)$ Rational Reduction & 200-Digit `mpmath.iv` Interval Enclosure:** Pure rational arithmetic over $\mathbf{Q}(i\omega_c)$ ($i^2\omega_c^2 = -2/13$, $e_c = 135/26$) proves with zero floating-point operations on the v1.3/v1.4 audit slice that $\|q_{raw}\|^2 = 13009/585$, $\mathrm{Re}(G_{21})/2 = 305328816/1498181485 > 0$, and $\ell_{1,v1.4} = (305328816 / 1498181485)\sqrt{(13/2)} = +0.519589117351220268983393194780597940008459836048 43$. At $\varepsilon_H = \varepsilon_I = 0.01$, 200-digit `mp.iv` interval arithmetic certifies $H_{obs,v1.5}^* \in [8, 8 + 4\times10^{-694}]$ and $\ell_{1,v1.5} > +0.519589 > 0$.
- **Post-Hopf Poincaré, Liouville & Floquet Benchmark:** At $e = 1.20\, e_c$ ($\varepsilon = 0.05$), trajectories from 3 initial conditions in $B(e)$ converge to an attracting periodic orbit on $\Sigma = \{\gamma = 1.0, \dot{\gamma} > 0\}$ with period $T = 18.957047$ s, `LSODA`/`DOP853` agreement $\|\Delta y\|_2 < 10^{-6}$, Liouville divergence integral $\int_0^T \mathrm{div}\, F\, dt = -10405.41$ ($\det(DP) \approx 10^{-4519} \ll 1$), and $\rho_{FD}(DP) < 10^{-8}$ at rtol = 10^{-12}.
- **Gate 0 Boundary:** Proposition 1 formulates a conditional covariant holographic dictionary ($H_{max} = \mathrm{Area}(\Sigma_{HRT})/(4\ell_P^2 \ln 2)$) for causal-diamond subsystems (`CONDITIONAL_FRONTIER`), while general non-geometric Gate 0 remains `OPEN`.

Gate width $\varepsilon_H = \varepsilon_I = \varepsilon$	Processing gate 1 − $\tanh(\dot{I}^*/\varepsilon)$	Jacobian shift $\|J_{v1.5} - J_{v1.4}\|_F$	Hopf threshold shift $\|\Delta e_c\|$	First Lyapunov shift $\|\Delta \ell_1\|$
ε = 0.10 (100-digit)	4.122307×10^{-9}	2.205936×10^{-8}	2.140429×10^{-8}	1.747892×10^{-6}
ε = 0.05 (100-digit)	8.496709×10^{-18}	4.546755×10^{-17}	4.411714×10^{-17}	7.207807×10^{-15}
ε = 0.02 (150-digit)	7.440152×10^{-44}	3.981379×10^{-43}	3.863156×10^{-43}	1.577807×10^{-40}
ε = 0.01 (750-digit)	2.767793×10^{-87}	1.481353×10^{-86}	1.437373×10^{-86}	1.107931×10^{-82}
ε = 0.005 (1450-digit)	$3.830339 \times 10^{-174}$	$2.049704 \times 10^{-173}$	$1.988999 \times 10^{-173}$	$6.112841 \times 10^{-169}$

OSF DOIs: 10.17605/OSF.IO/TSW4B | 10.17605/OSF.IO/NSEUG | 10.17605/OSF.IO/U7M6H | 10.17605/OSF.IO/S7WHX

Zenodo Appendix A v1.5 DOIs: 10.5281/zenodo.23497142 | 10.5281/zenodo.23501526 | 10.5281/zenodo.23534449
Repository & CI: github.com/reggaesharkk/physics-v1-dynamical-sector (CI Run #39587627124, VRK Digest `5eb316f25edef8e7...`)

6. Program 04: LRSC Odd-Ring Spectral Rank & Exact Finite Subset Certificates

Low-Rank Spectral Compression (LRSC) maintains a strict separation between two distinct layers: (1) an analytic single-step channel-space rank theorem for near-Nyquist odd cyclic rings, and (2) finite computer-assisted subset-cardinality certificates ($K_{0.001}$) for specific $M = 99$ benchmarks.

6.1 Frozen analytic odd-ring rank theorem lineage (rank(V) = p)

- **v1.2 Theorem (10.17605/OSF.IO/NM5BW):** For odd $M \geq 9$, near-Nyquist $k = (M - 1)/2$, carrier $X_t = 1 + A\cos(2\pi kt/M)$ ($A > 0$), local shedding $j_t = X_t - \min(X_{t-1}, X_t, X_{t+1})$, and any reflection-symmetric binary mask with at least one inactive reflection orbit and p active reflection orbits, the conjugate-balanced post-stencil channel matrix satisfies rank(V) = p. The proof combines residue-class Fourier nonvanishing ($\hat{j}_r \neq 0$ for $M = 4L + 1$ and $M = 4L + 3$), Chebyshev/Vandermonde orbit independence, and stencil injectivity on the masked span.
- **v1.2.1 & v1.2.2 Additive Supplements:** v1.2.1 adds proof clarification and finite corroboration across 197 odd rings ($M = 9..401$, 20,094 coefficients, max DFT error $< 2 \times 10^{-14}$). v1.2.2 additively records errata (correcting the derivative bracket to $f'(x) = -\sin(x)[\cos(2x) + 4\cos^2(x)]$ and noting $M = 9$ as the first admissible $4L + 1$ ring without altering the proof) and verifies 500,494 Fourier coefficients across $M = 9..2001$ plus 16,343/16,343 exhaustive masks on $M = 9..25$.

6.2 Exact finite $M = 99$ threshold certificate ($K_{0.001} = 11$)

Certified finite claim: For $M = 99$, $k_{idx} = 49$, $\theta = 0.08$, $A = 0.50$ ($N_{act} = 81$, $p = 41$, rank(V) = 41), the smallest subset cardinality achieving relative complex ℓ_2 error $\leq 10^{-3}$ is exactly $K_{0.001} = 11$ (giving the compression ratio $r_{LRSC} = 11/99 = 1/9$ used as a fixed parameter scale in Program 08).

Cardinality K	Enumerated subsets	Minimum integer lower score ($B = 10^{16}$)	Margin over pass threshold U = 5,045,156,716	Direct float64 relative error
1	50	2,952,642,845,575	+2,947,597,688,859	0.0241917969 (FAIL)
4	230,300	31,540,087,314	+26,494,930,598	0.0025003115 (FAIL)
7	99,884,400	15,193,292,406	+10,148,135,690	0.0017353561 (FAIL)
8	536,878,650	8,560,399,866	+3,515,243,150	0.0013025959 (FAIL)
9	2,505,433,700	7,143,182,466	+2,098,025,750	0.0011898947 (FAIL)

Cardinality K	Enumerated subsets	Minimum integer lower score ($B = 10^{16}$)	Margin over pass threshold U = 5,045,156,716	Direct float64 relative error
10	10,272,278,170	5,454,959,195	+409,802,479	0.0010398206 (FAIL)
11 (Witness S_{11})	Certified witness	[467259377740069839982565106 24, ...826] / 10^{35}	Strictly below $10^{-6}c$ lower bound	**0.0009623691 (PASS)**

- **Total nonempty subsets visited through $K = 10$:** 13,432,735,555, using outward integer rounding of trigonometric/DFT quantities with zero floating-point operations in the certified interval generator.
- **Passing witness:** S_{11} = {0, 1, 3, 42, 43, 44, 45, 46, 47, 48, 49}, stable across $\theta \in [0.071206344, 0.088346807]$ and $A \in [0.475795, 0.529975]$.
- **Rank vs. cardinality across adjacent masks:** At $\theta = 0.07$ ($p = 42$, rank(V) = 42), $K_{0.001} = 14$ (certified over 527,046,644,056 subsets through $K = 13$); at $\theta = 0.08$ ($p = 41$, rank(V) = 41), $K_{0.001} = 11$; at $\theta = 0.09$ ($p = 40$, rank(V) = 40), $K_{0.001} = 12$. Adjacent masks differ by a rank-one update $\|V_{0.07} - V_{0.08} - h\beta^{\top}\| = 1.05 \times 10^{-49}$.

DOIs: 10.17605/OSF.IO/NM5BW | 10.17605/OSF.IO/QJFR3 | 10.17605/OSF.IO/JAK5X | **Repository:** github.com/reggaesharkk/lrsc-odd-ring-spectral-rank

7. Program 05: Navier–Stokes Finite-to-Continuum Bridge Audit

The Navier–Stokes program is a finite-mode audit and bridge-engineering program for the unforced periodic 3D incompressible Navier–Stokes equations, not a claimed solution to the Clay Millennium problem. It combines spectral identities, analytical falsifiers, prospective finite cutoff gates, 128/192-bit Arb interval-certified ODE chains, and an explicit continuum-transfer frontier.

7.1 WP1–WP18 structural identities, falsifiers & prospective K36 line

- **Verified identities & falsified closures:** Localized energy, enstrophy, and smooth Gevrey identities were verified to 4.14×10^{-15}. Energy-only concentration bounds were falsified by fixed-energy scaling $u_\lambda(x) = \lambda^{3/2}v(\lambda x)$; enstrophy-power monomial majorants face a scaling/integrability conflict ($p \geq 5/4$ spatial scaling vs. $p \leq 1$ temporal integrability); helicity is phase-blind on an exact 6-mode zero-helicity witness.
- **Triadic & orbit compression diagnostics:** Established the mode-grouped cancellation hierarchy $|T_N| \leq B_N \leq A_N$ and isolated the persistent coordinate-magnitude orbit [0, 3, 6] across $N_8 \to N_{11}$.
- **WP16 prospective K36 line & preserved N17 failure:** The N11-derived K36 coalition passed predeclared time-gate tests at N12, N14, N15, N16 (with post-hoc exit accounting at $t = 0.0023$), and N17 (seed `20260942`, grid 64, 520 proposals, mass exit at $t = 0.0024$). The separate frozen N17 ordered-orbit/normalizer mechanism gate failed/became unevaluable at directional split ((1, 7, 9), (2, 7, 15)) and is preserved without retuning.

7.2 Finite N11 K36 crossing certificate & WP19 same-datum chain (N = 11,...,18)

For one unchanged 112-pair rational initial datum (witness SHA-256 `4789e27170f28279b3c6878f8874547b303d5cbd4a20848f3d5d8088bd10a624`, K36 key SHA-256 `7da5fc6d39ee03140d42ba40c5158cc20b043de33e4cc7f`

3ea52b71b79143f47), viscosity ν = 0.1, time horizon T = 0.003, and observable $F = I - 9O$, 120 whole-segment 128-bit Arb enclosures certify a positive-to-negative sign crossing across all eight finite Galerkin cutoffs N = 11, 12, 13, 14, 15, 16, 17, 18 with zero parameter retuning:

Cutoff / Package	Initial interval $F(0)$	Certified endpoint interval $F(0.003)$	Whole-path normalizer lower bound	Terminal error radius / Archive SHA-256
N11 Validated Turnover	[645.8037741471, 645.8037741472]	[−54.748409847, −42.032667894]	> 48,990.29795521	120/120 Arb; SHA-256 d29224e1dd4a...cfbe
N12 Same-Datum	[645.8037741471, 645.8037741472]	[−73.633221010, −70.396895629]	> 49,091.85228719	120/120 Arb; post-audit replay verified
N13 Same-Datum (v1)	[645.8037741471, 645.8037741472]	[−87.154087422, −83.563901281]	≥ 48,869.38355689	≤ 0.000012244529; SHA-256 858eeec4a1cb...ac5e
N14–N18 Chain (WP19 v0.21)	[645.8037741471, 645.8037741472]	N18: [−88.699660529, −84.159654865] (all 8 cutoffs ≤ −42.0327)	Family min > 48,850.68586052 (N18: 48,860.5957)	Max terminal error 0.000013774643 (at N18)

7.3 WP19 v0.16–v0.28 adjoint audit, 3 October M15 remainder test (PR #155) & continuum frontier

- **BGT critical-space gate (WP19 v0.16):** A 160-bit Arb lower gate proved that the generic Brunk–Giesselmann–Tscherpel sufficient criterion exceeds 1 on the first N14 Hermite segment (log LHS > $6.217838664529 \times 10^{12}$), ruling out generic-constant Gronwall certification for this state.
- **Terminal gradient intervalization & v0.28 backward-diffusion sign correction (1 Oct):** WP19 v0.27 certified the terminal signed-C500 degree-7 polynomial and analytic gradient at 128-bit Arb for 14 → 15 through 17 → 18 (width/prediction < 10^{-12}). On 1 October (PR #149/#150), WP19 v0.28 corrected an earlier reverse-diffusion note: in reverse time $\tau = T - t$, $e_\tau = -\mathrm{VJP}(u, e) - \nu\Lambda e + r$, so viscosity is dissipative in the propagated ℓ_2 error-radius recurrence under imported segment bounds.
- **3 October WP19 targeted M15-support remainder audit (PR #155):** A 192-bit Arb M15-support replacement shard across high-value steps 220–239 reduced the selected-step quadratic upper bound by **0.533%** (79,714,789,407,812.3283 → 79,290,095,908,202.2205) and the whole-path conservative remainder by **0.259%** (164,063,087,678,950.0886 → 163,638,394,179,339.9808). Because the signed integral lower bound is +5,758,574,435.60 ≈ 5.7586×10^9 while the scalar remainder is ≈ 1.6364×10^{14}, the lower signed margin remains negative — proving that global scalar majorants cannot close the bridge without modewise/vector-state control.
- **Continuum bridge frontier:** While direct newly opened shell contributions are controlled by boundary-annulus localization and `RadiiPolynomial.lean` (SHA-256 `f835b62b...`) formalizes the finite radii-polynomial discriminant, continuum closure requires cutoff-uniform control of recursive backreaction drift $\Gamma_M(P_M u_{M+1}) - \Gamma_M(u_M)$ and high-frequency vorticity continuation, which remain **OPEN**.

Zenodo DOIs: 10.5281/zenodo.22932635 (v0.1) | 10.5281/zenodo.22939984 (v0.2) | 10.5281/zenodo.23034545 (N13) | 10.5281/zenodo.23474505
Repository: github.com/reggaesharkk/navier-stokes-bridge-audit

8. Program 06: IntentSeal + Verifiable Research Kernel (VRK v0.3.1) / AgentOS Lineage

This program packages the portfolio's verification discipline into deterministic software infrastructure: claim-bound evidence schemas, fail-closed promotion gates, release hashes, deterministic certificates, and bounded scientific demonstrators.

Artifact / Layer	Key technical & verification evidence	Persistent identifier / digest
IntentSeal Verification Kernel (v0.1.1–v0.5.0 Freeze)	Preserves historical AgentOS Verification Kernel releases byte-for-byte; Linux test suites pass `10/10, 14/14, 34/34, 47/47, 61/61`; 161,324 bytes; 11 prototype limitations documented.	Zenodo 10.5281/zenodo.23016445 SHA-256: `05ba923c...ae5d4c0`
Verifiable Research Kernel (VRK v0.3.1 Final)	Separates proposition, evidence, dependency DAG, epistemic status (`intentseal-vrk-claim-v2`), and publication authority; 177,783 bytes; trust root at commit `e3b43046...`.	Zenodo 10.5281/zenodo.23018459 SHA-256: `c18c4c69...575cd79`
VRK Validation & Red-Team Suite	`30/30` unit/integration test suites pass; `33/33` adversarial claim, evidence, promotion, trust, and authority mutations rejected fail-closed.	Cert digest: `09209816...c597a8f` Trust root: `f010a99d...05a0d0f`
Demonstrator 001 & Demonstrator 002	Dem. 001: Finite LRSC $M = 99$, $\theta = 0.07$, $K_{0.001} = 14$ (527,046,644,056 subsets through $K = 13$). Dem. 002: Finite N11 K36 crossing package (tagged `EVIDENCE_BOUND`).	Ecosystem Zenodo: 10.5281/zenodo.23474505

Scope boundary: VRK is a deterministic proof-and-manifest workflow verifier, not a universal truth oracle or production security boundary. Historical AgentOS architectural documents (zero-trust policy engine, loop detector, bounded micropayment client) are system designs unless backed by the maintained IntentSeal/VRK implementation and tests.
Repositories: github.com/reggaesharkk/intentseal | github.com/reggaesharkk/Reggae-shark-universe-

9. Program 07: Riemann Bandwidth-One Verification & Analytic Zero-Proportion Track (WP84)

This program investigates the proportion of nontrivial zeros of $\zeta(s)$ that are simultaneously simple and lie on the critical line $\mathrm{Re}(s) = 1/2$, while strictly separating proved finite/operator theorems and exact polyhedral certificates from conditional model consumers and open arithmetic transfer problems.

9.1 External baseline vs. internal conditional and model numbers

Object / Figure	Evidence class	Exact mathematical meaning & claim boundary
> 67.25% (2/3 baseline)	Established External Literature	Published 2026 unconditional simple critical-line zero baseline (Alpoge–Furman `arXiv:2608.13637`; Lamzouri `arXiv:2609.02882`; Biao Wang $+6.666 \times 10^{-8}$ refinement `arXiv:2609.24167`).
68.82027318675% ($R = 400$ consumer)	Conditional Frontier	Author's $R = 400$ exact conditional consumer ledger (defect $\le 0.155898634 + (593/2000)\Delta_4 + \ldots$). Requires unpaid filtered-to-sharp fourth-order correction and Weil transfer; **not** an unconditional bound.

Object / Figure	Evidence class	Exact mathematical meaning & claim boundary
79.6270966813% (11061/13891)	Conditional Model Value	Exact degree-3 Christoffel polynomial optimization $P(x) = 1 - (43428/13891)x + (37704/13891)x^2 - (9660/13891)x^3 \Rightarrow w_0 = 1415/13891$, $1 - 2w_0 = 11061/13891$ on imported reference moments (1, 1, 4/3, 2, 13/4, 101/18, 12809/1260), with bounded rational ladder $F_\delta(x) = (\delta + 1)P(x)^2 / (P(x)^2 + \delta)$ ($S(61) = 79.2931\% \to S(18{,}577) = 79.6260\%$). Model identity only; **not** a theorem about actual $\zeta(s)$ zeros.
Strict > 79% & RH	Open Frontier	Explicit project goal at $R = 6000$ ($A = 11501/12500$, $B = 137641/250000$, needed signed net saving > 0.002057355). Joint signed 3rd/5th/6th moment inequality, prime-to-Weil correspondence, zero-side transfer, and RH remain **OPEN**.

9.2 Specialized proved & computer-certified results (Public Release v0.1.0)

- **A. Curvature-Controlled Discrete Loewner Band Theorem (`PROVED / ANALYTIC`):** For an $n \times n$ real-symmetric matrix with off-diagonal entries $H_{ij} = (\alpha_i - \alpha_j)/(\pi(j - i))$, band truncation H_R ($|i - j| \le R$), $q = \|\alpha\|_2^2/n$, and normalized trace $\tau = \mathrm{tr}/n$, any real C^2 function f with bounded second derivative satisfies $|\tau f(H) - \tau f(H_R)| \le 12q\|f''\|_\infty / (\pi^2 R)$ for $R \ge 1$ (combining 4 + 8 = 12 from quadratic Taylor trace remainder and commutator divided-difference bounds; standalone matrix theorem independent of $\zeta(s)$).
- **B. Exact C_5 Connected Five-Cycle Polyhedral Sign & Support Certificate (`CERTIFIED`):** For the 150-term five-cycle connected kernel $C_5(v)$ on the closed outer-support polytope with half-L^1 norm $s(v) = \frac{1}{2}\sum_{j=1..5} |v_j|$, exact rational evaluation over all C(45, 4) = 148,995 hyperplane intersections (`151` surviving support vertices, checked across 16 CI shards) proves $0 \le C_5(v) \le \max(s(v) - 1, 0)$ and $C_5(v) = 0$ whenever $s(v) \le 1$ (vertex min 0, max 1/2, model integral 1/36).
- **C. Filtered Fourth-Cumulant Sign, Indefinite-Gram & Pointwise Domination Counterexamples (`CERTIFIED / FALSIFIED`):** At $R = 400$, $v = (1/5, -2/5, 3/5, -2/5)$ ($5 \mid m$), the filtered Fourier cumulant is enclosed in [0.39974848084713831, 0.39974848084713832], and a 2 × 2 order-averaged connected pair kernel has cross entry ≈ 0.39950281400985 and negative determinant $-c^2$, refuting universal pointwise/Gram positivity shortcuts. Separately, at $x = 4/5$, the shifted consumer $p_s(4/5) \approx 0.115178 > F_{61}(4/5) \approx 0.014630$, falsifying pointwise domination $p_s \le F_{61}$, and the order-4 consistency ledger enforces $t_{adj} = 7/60$, $t_{opp} = 1/30$, $\Phi_4 = -1/60 \Rightarrow m_4 = 1/4$ (rejecting memoryless 7/30 and 4/15 closures).
- **D. 9-Site Gaussian Loewner Certificate & Source-Free Dyadic Replay `A1294` (`CERTIFIED`):** Exact Wick/pinching sixth-moment lower bound via two independent finite enumerators (`531,441` walks; `40,191` edge multisets), and standalone Python standard-library (`fractions.Fraction`) replay of finite dyadic certificate `A1294` ($A = 647/50000$, $B = 1094977/5000000000$, visiting 13,454,121 tree nodes and 6,727,061 terminal leaves; archive SHA-256 `11c1ab182003...cd19`, manifest SHA-256 `961a0281a134...2a5b`).

Public Release & CI Provenance (8 Oct 2026): Curated 38-file snapshot tagged `v0.1.0` at commit `6d799bddd9a27ba79c4faf8c757f7985ceb6d4c2`; GitHub Actions Run #37810408337 passed **21/21 jobs**.
Zenodo DOIs: 10.5281/zenodo.23244818 (Version) | 10.5281/zenodo.23244817 (Concept) | **Repository:** github.com/reggaesharkk/riemann-bandwidth1-verification

10. Program 08: The Theory of Bounded Existence — 4D Soft-Wall ODE System (Appendix B v1.0)

Released on 24 October 2026 (DOI: 10.5281/zenodo.23608244), **Appendix B v1.0** extends the 3D soft-wall ODE of Program 03 into a dimensionless, unit-normalized 4D system $x(t) = (\gamma(t), H_{obs}(t), \dot{I}(t), \Psi(t))$ ($a = b = c = f = g = E_{avail} = 1$, $\gamma_{max} = 10$, $H_{max} = 100$, $\dot{I}_{erase} = 10$, $\varepsilon_H = \varepsilon_I = \sigma_0 = 0.05$, $\mu_\Psi = 0.60$, $r_{LRSC} = 1/9$) coupling the self-modeling readout $\Psi(t)$ (Program 02) and spectral ratio $r_{LRSC} = 11/99 = 1/9$ (Program 04) into the effective drive $P_{eff}(t) = P_0 + \Lambda_{eff}\gamma(t) + \lambda_{lucid}\Psi(t)$ ($\Lambda_{eff} = \lambda_{sim} r_{LRSC}$) with smooth ($C^\infty$) quadratic self-feedback ratio $R_{self}(s) = \mathrm{fb}(s)^2 / [\mathrm{fb}(s)^2 + (fP_0)^2 + \sigma_0{}^2]$, where $\mathrm{fb}(s) = e \tanh(\dot{I}/\varepsilon_I)\, \dot{H}_{obs}$.

10.1 Exact Hopf monotonicity, block-triangular J_4 decoupling & 17-theorem Lean 4 formalization

- **Exact Rational Hopf Locus & Monotonicity ($de_c/dP_0 > 0$ on $P_0 > 10/99$):** In the $\varepsilon_H, \varepsilon_I \to 0^+$ limit on the interior positivity domain $P_0 > P_{crit} = 10/99 \approx 0.10101$ (where $H_{obs,v1.4}{}^* = 100 - (10 + P_0)/P_0 > 0$), $e_c(P_0) = P_0(P_0 + 20)(11P_0 + 10)(P_0{}^2 + 120P_0 + 100) / [100(P_0 + 10)(P_0{}^2 + 130P_0 + 200)]$ yields exact rational thresholds $e_c(5) = 377/84 \approx 4.488095$, $e_c(4) = 12069/4025 \approx 2.998509$, $e_c(1) = 97461/364100 \approx 0.267676$, and strictly positive derivative $de_c/dP_0 > 0$ (all 8 primitive numerator coefficients `[11, 3085, 250000, 5542000, 49530000, 120550000, 94000000, 20000000]` and 7 denominator coefficients strictly positive).
- **Block-Upper-Triangular J_4 Decoupling:** At any steady state ($\dot{H}_{obs}{}^* = 0$), both $R_{self}{}^* = 0$ and $\nabla R_{self}|_* = (0, 0, 0, 0)$ vanish by quadratic construction, so row 4 of J_4 is $[0, 0, 0, -\mu_\Psi]$ and $\det(\lambda I_4 - J_4) = (\lambda + \mu_\Psi) \det(\lambda I_3 - J_3(\Lambda_{eff}))$ for all $\lambda_{lucid} \geq 0$, yielding the closed-loop Routh–Hurwitz threshold $e_c{}^{CL}(P_0, \Lambda_{eff})$.
- **17 Machine-Checked Lean 4 Pointwise Theorems (`ExistenceVerification.lean`):** Verified in GitHub Actions CI Run #6 (Commit `c690801`, 6m46s on `main`) via `lake build BoundaryVerification Existence Verification` (`Build completed successfully.`, `#print axioms` → solely `[propext, Classical.choice, Quot.sound]`, zero `sorryAx`, zero `nlinarith`). Proves `R_self_sq_nonneg`, $-M_H \leq \dot{H}_{obs} \leq M_H$, the linear comparison lower bounds `dgamma_dt_4D_ge_neg_decay` ($\dot{\gamma} \geq -b\gamma$) and `dIdot_dt_4D_ge_linear_floor` ($\ddot{I} \geq fP_0 - (eM_H/\varepsilon_I + g)\dot{I}$), all 8 face inequalities on $B_4 = [0, \gamma_{max}] \times [0, H_{max}] \times [0, \dot{I}_{max,4D}] \times [0, 1]$ (including `dIdot_dt_4D_nonneg_at_boundary_zero_drive` for $P_0 \geq 0$), and `existence_box_8face_nagumo_certificate`. Classical Nagumo and ODE comparison theorems then give $x(t) \in B_4$, $\gamma(t) \geq \gamma(0)e^{-bt} > 0$, and uniform positive floors $\liminf_{t\to\infty} \gamma(t) \geq \gamma_{floor}(P_0) > 0$ for $P_0 > 0$.

10.2 Six dynamical regimes, 2 × 2 zero-drive ablation & 20D variational monodromy audit

At $P_0 = 0$ and $0 \leq \Lambda_{eff} < 1$, the equilibrium quadratic has roots $\gamma^* = 0$ and $\gamma^* = -10(1 - \Lambda_{eff})/\Lambda_{eff} < 0$ (-2.5 at $\Lambda_{eff} = 0.80$), so **no positive interior equilibrium or Hopf bifurcation exists** (`None (No Int. Eq.)`), and the origin (0,0,0,0) is locally asymptotically stable (eigenvalues $\{-0.105573, -1.894427, -200.0, -0.60\}$ at $\Lambda_{eff} = 0.80$). A 2 × 2 ablation of $(\lambda_{sim}, \lambda_{lucid})$ at $P_0 = 0$ demonstrates that the memoryless proportional loop alone ($\lambda_{sim} = 7.20$, $\lambda_{lucid} = 0$) collapses to the origin ($\|x(400)\|_2 = 1.84 \times 10^{-16}$) because $\gamma_{min} \approx 3.6 \times 10^{-4}$ at the erasure trough, whereas the slow Ψ-memory reservoir ($\mu_\Psi = 0.60$, $\Psi_{min} = 0.7305 \Rightarrow \lambda_{lucid}\Psi \geq 0.219$) bridges the trough and sustains a bistable relaxation cycle coexisting with the locally stable origin. Forward bisection in λ_{lucid} from $x_0 = (2.05, 95.0, 4.0, 0.05)$ locates the basin-dependent **cycle-**

persistence threshold from x_0 at $\lambda_{\text{lucid,crit}}(x_0; 0) = 0.080341$ ($\Lambda_{\text{eff}} = 0$) and $\lambda_{\text{lucid,crit}}(x_0; 0.80) = 0.057974$ ($\Lambda_{\text{eff}} = 0.80$, a −27.8% reduction from x_0).

Regime / Slice (e = 3.60 on I–IVc)	(P_0, λ_{sim}, λ_{lucid}, κ_{meta})	Threshold e_c^{CL}	Period T (s)	Period avg $\bar{P}_{\text{eff},T}$	Peak max Ψ	3-IC Δ_{IC} & Variational Floquet
I. Waking Fixed Point	(5.0, 0.0, 0.0, 0.20)	4.4881	Fixed Pt	5.000000	0.0000	Stable eq. ($\gamma^* = 10/3$, $\Psi^* = 0$)
II. Low-Ψ Cycle	(4.0, 0.0, 0.0, 0.20)	2.9985	19.685200	4.000000	0.1726	$\Delta_{\text{IC}} = 2.58\times10^{-11}$; $\mu_\Psi = 2.54\times10^{-6}$
IIIa. High-Ψ Readout	(4.0, 0.0, 0.0, 2.50)	2.9985	19.685200	4.000000	0.7689	Decoupled readout (ΔT = 0.0 s vs. II)
IIIb. Coupled High-Ψ	(4.0, 0.0, 0.30, 2.50)	2.9985	19.520419	4.229492	0.7706	$\Delta_{\text{IC}} = 5.61\times10^{-11}$; $\rho_{\text{var}} < 10^{-8}$
IVa. Autopoietic (P_0 = 1)	(1.0, 7.20, 0.30, 2.50)	1.3938	18.149080	3.266554	0.8022	$\Delta_{\text{IC}} = 6.23\times10^{-11}$; $\rho_{\text{var}} < 10^{-8}$
IVb. Zero-Drive (P_0 = 0)	(0.0, 7.20, 0.30, 2.50)	None (No Int. Eq.)	19.387103	2.190321	0.8124	Bistable ($\Delta_{\text{IC}} = 3.41\times10^{-11}$; $\rho_{\text{var}} < 10^{-8}$)
IVc. Ψ-Only (P_0 = 0)	(0.0, 0.0, 0.30, 2.50)	None (No Int. Eq.)	17.936755	0.239590	0.8125	Bistable ($\Psi_{\text{min}} = 0.7305$; $\rho_{\text{var}} < 10^{-8}$)
Min Slider Corner	e = 0.1, P_0 = 0.1, κ = 0.05	0.099132	32.559724	0.100000	0.0155	Soft-wall H_{sw}^* = +0.1197 bits > 0
Max Slider Corner	e = 6, P_0 = 6, λ_{sim} = 10, λ_{luc} = 1	21.958109	Fixed Pt	12.078582	0.0000	Exact $\gamma^* = (-11 + \sqrt{1471})/5 = 5.470724$

Scope firewall: "Self-sustaining" denotes a positive-feedback loop with sigmoidal saturation generic to relaxation oscillators; the 9.14× increase in $\bar{P}_{\text{eff},T}$ between IVc (0.239590) and IVb (2.190321) is an internal property of $\Lambda_{\text{eff}}\gamma(t)$ in this ODE, with no physical energy accounting, thermodynamic claim, or clinical sleep-stage equivalence.

Zenodo DOI: https://doi.org/10.5281/zenodo.23608244 (CC BY 4.0 / MIT) | **Repository & CI:** github.com/reggaesharkk/theory-of-bounded-existence (CI Run #6, Commit `c690801`, Digest `80e0724e98e68de2...`)

11. Program 09: STRENGTH H4-BRIDGE-1 Audit & Seven-Problem Attack Board

11.1 Mathematical & empirical investigation of H4-BRIDGE-1 high-noise J75 failure (13 October 2026)

Audited the preserved `H4-BRIDGE-1-v1` (100 networks, 600 delay rows) and `H4-BRIDGE-0-v1` (36 networks) ledgers across `RNN16` (d = 16, 338 parameters) and `GRU9` (d = 9, 344 parameters) to determine why high-

noise joint training `J75` ((w_T, w_M) = (0.75, 0.25) at state-noise SD σ = 0.25) failed its predeclared 8/10 engineering gate (4/10 marker pass, 9/10 task pass, 4/10 joint pass):

- **Analytical BPTT Exactness Verified (`H_BPTT_BUG` Refuted):** Independent derivation and central finite-difference testing across all 338 (`RNN16`) and 344 (`GRU9`) parameters confirmed exact analytical derivatives ($\max|\nabla_{FD}^{(2)} - \nabla_{analytic}| = 1.43 \times 10^{-10}$ for `RNN16`, 1.62×10^{-10} for `GRU9`; 4th-order Richardson error ≤ 4.15×10^{-13}).
- **Four Interacting Failure Mechanisms Isolated:** (1) *$D - 2$ Path-Length Invariance & Spurious Null Noise Floor:* While raw $\|g_M\|_2$ stays constant (≈ 0.18925 in `RNN16`, ≈ 0.06005 in `GRU9`), over 99.99% at $D \in \{32, 64\}$ is the zero-marker null floor $\|g_M^{null}\|_2$, whereas the causal signal $\|g_M^{signal}\|_2$ decays across $D - 2$ Jacobian multiplications from 0.07049 (`D8`) → 0.00301 (`D16`) → 7.0×10^{-6} (`D32`) → 1.6×10^{-10} (`D64`) in `RNN 16` (and 0.02432 → 3.6×10^{-7} in `GRU9`); (2) *Adam Shared Second-Moment v_k Clamping:* In joint training, the $O(w_T)$ task gradient from the final 16-step window $S_3 \in [48, 64)$ clamps Adam's shared $\sqrt{\hat{v}_k}$, destroying the adaptive rescaling that allows `Arm M` ($w_T = 0$) to pass 10/10; (3) *Short-Horizon Task Sufficiency vs. Capacity Antagonism:* By Sheppard's orthant theorem for Gaussian block sums, reading S_3 alone achieves ½ + (1/π) arcsin(1/√3) = 69.59% task accuracy and $S_2 + S_3$ achieves 80.41%, driving recurrent eigenvalues into a leaky regime (α ≈ 0.85–0.92) antagonistic to 64-step bistable latching (which collapses `GRU9` linear memory `mc64` by 88%, from 0.926 to 0.113); and (4) *Unweighted Head L2 Non-Superposition & `measure_bridge1.py` Lag Indexing Bug:* Output heads `wy, wm` received unweighted L2 penalties (amplifying marker head shrinkage 4× in `J75`), and `measure_bridge1.py` read 1-based step t_{read} = lag for a pulse placed at t_0 = 64 − lag (reading before the pulse occurred on `D1..D32`).
- **Prospective H4 Status:** Formulated a leak-free Leave-One-Seed-Block-Out (LOSO) cross-validation and 10,000-permutation Freedman–Lane block test ($M \geq 20$ master seeds) with PCGrad / decoupled Adam moments and in-channel interventional perturbations (`IRP`). Confirmatory H4 incremental predictive value remains **OPEN**.

Family & Delay D (σ = 0.25)	**Task Grad $\|g_T\|_2$**	**Raw Marker $\|g_M\|_2$**	**Null Floor $\|g_M^{null}\|_2$**	**Causal Signal $\|g_M^{signal}\|_2$**	**Causal SNR $\|g_M^{sig}\|/\|g_M^{null}\|$**	**Ch-0 Input $\|g_{M,ch0}^{sig}\|_2$**
`RNN16` (338 params) — `D8 / D16`	0.2184 / 0.2203	0.1991 / 0.1894	0.18925	0.070486 → 0.003014	0.3725 → 0.0159	0.009786 → 0.000263
`RNN16` (338 params) — `D32 / D64`	0.2203 / 0.2203	0.18925 / 0.18925	0.18925	7.0×10^{-6} → 1.6×10^{-10}	4×10^{-5} → 8.5×10^{-10}	4×10^{-7} → 1.1×10^{-11}
`GRU9` (344 params) — `D8 / D16`	0.0642 / 0.0640	0.0626 / 0.0601	0.06005	0.024323 → 0.004967	0.4050 → 0.0827	0.009639 → 0.001240
`GRU9` (344 params) — `D32 / D64`	0.0640 / 0.0640	0.06005 / 0.06005	0.06005	1.37×10^{-4} → 3.6×10^{-7}	0.0023 → 6.0×10^{-6}	2.8×10^{-5} → 5.4×10^{-8}

11.2 Seven-Problem Attack Board v0.6 (3 October 2026)

The Seven-Problem Attack Board is an explicit research and falsification agenda mapping entry gates and obstructions across seven major open problems. It enforces a falsifiable milestone discipline and is never

represented as a list of solved problems:

Target Problem	Current Portfolio Foothold & Audited Evidence	Formal Frontier Status & Prohibition
3D Navier–Stokes	Finite Galerkin $N_{11} \to N_{18}$ Arb chain, WP19 v0.28 adjoint sign fix, PR #155 M15 remainder audit, Ra `diiPolynomial.lean`.	**ACTIVE / OPEN:** Do not claim continuum 3D global regularity or blow-up.
Riemann / Zeta Zeros	Discrete Loewner $12/(\pi^2 R)$ bound, C_5 151-vertex polyhedral certificate, R = 400 & 11061/13891 conditional models.	**ACTIVE / OPEN:** Do not claim RH or an unconditional 68.82% / 79.63% record.
P versus NP	LRSC exact 13.43×10^9 & 527×10^9 branch-and-bound subset exclusion certificates used as methodological precedent.	**FORMALIZATION ONLY:** Barrier mapping; no complexity-class separation claimed.
Yang–Mills Mass Gap	Cutoff-uniform transfer target framing analogous to WP19 recursive drift bounds.	**FORMALIZATION ONLY:** Attack outline; no 4D QFT construction or mass gap proof.
Quantum Gravity	3D/4D soft-wall phenomenological ODEs (Programs 03 & 08) and conditional HRT Gate 0 dictionary (Proposition 1).	**GATE 0 OPEN:** No microscopic spacetime derivation or non-geometric bridge.
Dark Matter & Dark Energy	3-state (R, Ψ, Γ) cosmology sandbox equilibrium & internal falsifiers ($\Pi = 1$ contraction pitfall).	**HYPOTHESIS DESIGN:** Toy ODE falsifiers only; no physical dark-sector model claimed.

12. Cross-program architecture of verification

1. **Freeze the question before the decisive data or holdout** whenever the design is confirmatory or prospective.
2. **Bind every important artifact to a SHA-256 hash, manifest, deterministic seed, Git commit, or DOI deposit.**
3. **Use independent implementations and solvers** (e.g., Python vs. C++ enumerators; `LSODA` vs. `DOP853`; `SymPy` vs. `mpmath.iv`) whenever a claim could hinge on one code path.
4. **Prefer exact rationals or outward-rounded intervals** (`fractions.Fraction`, 128/192-bit `Arb`, 200-digit `mp.iv`) whenever a finite claim can be made rigorous.
5. **Separate theorem, finite certificate, simulation, diagnostic, conditional model, and open problem** in all public language via explicit `scope.does_not_imply` firewalls.
6. **Preserve negative results and failed gates** (N17 mechanism gate, BGT criterion, $p_s \neg\leq F_{61}$, H4-BRIDGE-1 J75) as permanent first-class audit records.
7. **Keep finite-dimensional and continuum claims strictly separate**, especially in Navier–Stokes and Riemann WP84.
8. **Treat corrections as additive provenance events**, leaving historical frozen archives byte-intact.

13. Chronology and research progression

Period / Date (2026)	Documented milestone & archival event
Early–Mid Sep	Self-Referential Processing v8.5 → v8.6.3 protocol hardening; 1,000-seed calibration (6.3% null reject); §6.4 API snapshot audit; OSF preregistration (10.17605/OSF.IO/37H2Y).

Period / Date (2026)	Documented milestone & archival event
17–24 Sep	Design Lab Phase 7 prospective simulation validation: 4,000 GLMM fits across 500 seeds; M1 selected over M2 under locked screen (10.17605/OSF.IO/NV7DZ); Ψ v4 benchmark archived (10.17605/OSF.IO/N3DQU, 8CFQW).
23–27 Sep	LRSC v1.2 analytic odd-ring rank theorem rank(V) = p (10.17605/OSF.IO/NM5BW), v1.2.1/v1.2.2 supplements, and exact M = 99 $K_{0.001}$ = 11 subset certificate (13,432,735,555 subsets, JAK5X).
24–28 Sep	Navier–Stokes WP1–WP18 identity audits, falsifiers, prospective K36 tests (N12–N17, preserving N17 mechanism failure), and N11 K36 128-bit Arb crossing certificate (10.5281/zenodo.22939984).
27–28 Sep	IntentSeal v0.1.1–v0.5.0 prototype freeze (10.5281/zenodo.23016445) and VRK v0.3.1 final release (30/30 tests, 33/33 adversarial rejections, 10.5281/zenodo.23018459).
29–30 Sep	WP19 same-datum finite chain certified through N18 (N = 11..18, max error 0.000013774643); v0.16 BGT critical-space gate; v0.27 128-bit Arb terminal signed-C500 gradient intervalization.
1–3 Oct	First 6-program Master Dossier compiled; WP19 v0.28 reverse-time viscosity sign correction (PR #149/#150); 3 Oct 192-bit Arb M15-support remainder audit (PR #155) and Seven-Problem Attack Board v0.6.
8–11 Oct	Riemann Bandwidth-One Verification v0.1.0 public release (38 files, 21/21 CI jobs in Run #37810408337, 10.5281/zenodo.23244818); 9 Oct 19-page Evidence Audit Dossier; 11 Oct Physics v1.4 scope note & ORCID 13-record index.
13–19 Oct	STRENGTH H4-BRIDGE-1 mathematical & empirical audit (refuting `H_BPTT_BUG`, isolating $D - 2$ causal SNR collapse & Adam v_k clamping); unified ecosystem release (10.5281/zenodo.23474505).
21 Oct	Physics Appendix A v1.5 Soft-Wall Completion: 6-face Lean 4 `BoundaryVerification.lean` (CI Run #39587627124, commit `4211c6f`), exact $\mathbf{Q}(i\omega_c)$ rational ℓ_1, 200-digit `mp.iv` enclosure (10.5281/zenodo.23534449).
24 Oct	Theory of Bounded Existence (Appendix B v1.0): 4D soft-wall ODE, 17-theorem Lean 4 `ExistenceVerification.lean` (CI Run #6, commit `c690801`), P_0 = 0 Ψ-memory bistability & 20D variational monodromy (10.5281/zenodo.23608244).

14. Current open frontier and theorem debt register

Program	Next rigorous mathematical / operational question	Do not claim yet (Global Claim Firewall)
01. Design Lab	Execute §6.4 API snapshot amendments without editing frozen v8.6.3; preregister a new protocol if adopting M1 GLMM.	No live empirical self-referential effect; no completed confirmatory study.
02. Ψ Benchmark	Run a preregistered multi-model confirmatory study if the v4 2AFC diagnostic is to support population inference.	No machine consciousness, subjective selfhood, or sentience conclusion.
03. Physics v1.5	Derive a microscopic/quantum-statistical origin for (α, β, γ, δ, μ, g) and close non-geometric Gate 0; prove global post-Hopf limit-cycle uniqueness analytically.	No general microscopic law of nature, quantum gravity solution, or cosmological dark-energy derivation.
04. LRSC	Explain analytically how minimum passing subset cardinality K_τ varies with mask threshold θ while preserving rank(V) = p.	No universal fixed-unit compression law or multi-step continuum PDE theorem.

Program	Next rigorous mathematical / operational question	Do not claim yet (Global Claim Firewall)
05. Navier–Stokes	Replace scalar remainder majorants with modewise/vector-state adjoint bounds (PR #155) and bound recursive backreaction drift $\Gamma_M(P_M u_{M+1}) - \Gamma_M(u_M)$.	No all-N persistence theorem; no 3D continuum Navier–Stokes global regularity or blow-up proof.
06. IntentSeal / VRK	Isolate gateway/attester/executor into separate OS principals and formalize a production security threat model.	No universal truth oracle or production security boundary guarantee.
07. Riemann WP84	Prove a signed 3rd/5th/6th moment saving at R = 6000, prime-to-Weil source normalization, and zero-side transfer.	No unconditional 68.82% or 79.63% zeta-zero record; no proof of the Riemann Hypothesis.
08. Bounded Existence	Perform numerical continuation of the unstable companion cycle at P_0 = 0 and interval-arithmetic (CAPD/Arb) enclosure of the 4D return map.	No thermodynamic free-energy generation, no clinical EEG sleep equivalence, and no metaphysical qualia claim.
09. STRENGTH H4	Run prospective $M \geq 20$ master-seed LOSO cross-validation with PCGrad / decoupled Adam moments and in-channel `IRP` pulses.	No confirmed incremental predictive value of `IRP` over `MC64/IPC` yet (H4 remains `OPEN`).

15. Master DOI, repository, CI run, hash, and reproducibility index

Program / Archival Artifact	Persistent Identifier (DOI), Repository, or Verified CI Run
Umbrella Master Repository	https://github.com/reggaesharkk/Reggae-shark-universe-
01. Self-Referential Processing v8.6.3 (OSF)	https://doi.org/10.17605/OSF.IO/37H2Y
01. Design Lab Consolidated & Phase 7 (OSF / GitHub)	https://doi.org/10.17605/OSF.IO/NV7DZ | github.com/reggaesharkk/self-referential-processing-designlab
02. Ψ Self-Modeling Organization & Benchmark (OSF / IA / GitHub)	10.17605/OSF.IO/N3DQU | 10.17605/OSF.IO/8CFQW | 10.17605/OSF.IO/CAE2H | github.com/reggaesharkk/psi-self-modeling-benchmark
03. Physics v1.0 & Appendix A v1.4 / Gate 0 (OSF)	10.17605/OSF.IO/TSW4B | NSEUG | U7M6H | S7WHX | 5MKET
03. Physics Appendix A v1.5 Soft-Wall (Zenodo / GitHub / CI)	10.5281/zenodo.23497142 | 23501526 | 23534449 | github.com/reggaesharkk/physics-v1-dynamical-sector (CI Run #39587627124)
04. LRSC Odd-Ring Rank & Lineage (OSF / GitHub)	10.17605/OSF.IO/NM5BW | G8XEY | QJFR3 | EW85B | JAK5X | github.com/reggaesharkk/lrsc-odd-ring-spectral-rank
05. Navier–Stokes Bridge Audit (Zenodo / GitHub)	10.5281/zenodo.22932635 (v0.1) | 22939984 (v0.2) | 23034545 (N13) | github.com/reggaesharkk/navier-stokes-bridge-audit
06. IntentSeal & VRK v0.3.1 (Zenodo / GitHub)	10.5281/zenodo.23016445 (IntentSeal) | 10.5281/zenodo.23018459 (VRK v0.3.1) | 23474505 (Ecosystem) | github.com/reggaesharkk/intentseal
07. Riemann Bandwidth-One Verification v0.1.0 (Zenodo / GitHub / CI)	10.5281/zenodo.23244818 (v1/v0.1.0) | 23244817 (Concept) | github.com/reggaesharkk/riemann-bandwidth1-verification (CI Run #37810408337)

Program / Archival Artifact	Persistent Identifier (DOI), Repository, or Verified CI Run
08. Theory of Bounded Existence Appendix B v1.0 (Zenodo / GitHub / CI)	https://doi.org/10.5281/zenodo.23608244 | github.com/reggaesharkk/theory-of-bounded-existence (CI Run #6, Commit c690801)

Appendix A. Key quantitative evidence & cryptographic hashes

A.1 Program 01 (Design Lab Phase 7 & v8.6.3) frozen hashes

- `cce97faba969a6da873d9a5a34d9da6421fdf24d1a04df29c1a599fb3044e3d0` — `phase7_results.csv`
- `e77755e76a3fa75ed8a130e578b864af75ca2b42a614f84306bc9b097f5e605e` — `phase7_summary.csv`
- `17c16f1279a72f2916a97544586e7a6095699281a3a21bcdfd3c0b21735ff1eb` — `phase7_paired.csv`
- `9ee3720597b35c553f0b3dd7a6f2ca159b904609b171cd41b4f0800ab2f8eae2` — `phase7_blocks_manifest.csv`
- `ae92741d95ce01c46b7b197cc7d18386a3ab87494fdf2c1f506682b270b6c318` — `phase7_integrity_report.txt`
- `7c33848597dd1fd928b600369ccf2ce11b483eeeedbf97f0e62b66ee8fcfce0d` — `phase7_design_lock.md`
- `128ff78bb6beb1640fa51897ab97dbb9d07dc56de276973d05a48dda524b53d3` — `items_pool_v1.json` (v8.6.3 genesis entry `a303cdeb8625...`)

A.2 Program 04 (LRSC) & Program 05 (Navier–Stokes) core numbers

- **LRSC $M = 99$ exact certificate:** Pass threshold integer upper bound U = 5,045,156,716 ($B = 10^{16}$); K = 10 global minimum integer lower score 5,454,959,195 (margin +409,802,479); K = 11 witness residual-square interval [467259377740069839982565106 24, 467259377740069839982565108 26] / 10^{35}; channel binary SHA-256 `9c6a4527716cf65a65f26887bbe76edb68b819f770c6a903f47a8e0d2e36bec9`.
- **Navier–Stokes finite chain & PR #155:** N11 archive SHA-256 `d29224e1dd4ad9f9454951415a3b080bc9f092839e24caaeddd056013785cfbe`; N13 archive SHA-256 `858eeec4a1cb323d23e91ffa9914af823a388a78ecc39c758abea3fcdec9ac5e`; $N_{11} \to N_{18}$ max terminal trajectory error 0.000013774643; min normalizer lower bound 48850.68586052; PR #155 M15-support whole-path conservative remainder 163,638,394,179,339.9808 vs. signed integral lower bound +5,758,574,435.60; `RadiiPolynomial.lean` SHA-256 `f835b62b1432f83134e7a8e5b4c136364bf21092a4e9e3e3b7b258cf09918239`.

A.3 Program 06 (IntentSeal/VRK) & Program 07 (Riemann WP84) core digests

- **IntentSeal & VRK v0.3.1:** IntentSeal prototype freeze SHA-256 `05ba923c90dfc481a70a3784b72d1cd71a0adeac2c34a22631169f545ae5d4c0`; VRK v0.3.1 release SHA-256 `c18c4c697d8188490e96530cfb5ddcb2e63f7c65aa560415401b1541e575cd79`; VRK deterministic certificate digest `09209816a034020076e0f66e1b8924b6dd1567c95f7159070f58ce62ac597a8f`; verifier trust root `f010a99d78b360b8fd5dc9b727f21b898889c6e75e3093cc22d6f04d005a0d0f`.
- **Riemann v0.1.0 & A1294 replay:** Git tag `v0.1.0` commit `6d799bddd9a27ba79c4faf8c757f7985ceb6d4c2` (CI Run `37810408337`, 21/21 jobs); A1294 certificate archive SHA-256 `11c1ab1820039284f60e492fe609f6a497175284acdd0c3c53d6ab62b8b1cd19` (13,454,121 nodes, 6,727,061 leaves).

A.4 Program 03 (Appendix A v1.5) & Program 08 (Appendix B v1.0) certificates

- **Shared Lean 4 Lake Toolchain:** `leanprover/lean4:v4.36.0-rc1` (`lean-toolchain` SHA-256 `964acf49a540...`); Mathlib rev `10156080d0e7997b71f3ef86bd6f0211b42ab90d` (`lake-manifest.json` SHA-256 `54ca7b4da415...`); `BoundaryVerification.lean` SHA-256 `49229b6163b5a3a77dc83779a7dc5350502ff2b3d6a7f2de4d39996f4460e417`.
- **Appendix A v1.5 (Program 03):** Exact rational $\ell_{1,v1.4}$ = (305328816 / 1498181485)√(13/2); VRK deterministic certificate digest `5eb316f25edef8e711db7ddec88224172a9cadd8cf7d332457dfa596f4368117`; Zenodo archive ZIP SHA-256 `05f73350776de86d2d16d65578f2575a5bf032fb8d97d153ba1fa215d79c5db2`.
- **Appendix B v1.0 (Program 08):** `ExistenceVerification.lean` (17 theorems, `0 sorry`, `0 nlinarith`) SHA-256 `eec9486803fb2e12f94c3ef966098acd31e4a66df8269ebe8cb59a937f7abcae`; `lakefile.lean` SHA-256 `bd0a126013dc34410b51a1527d02d96520df32a283e46cddb55db5acc0c6636c`; Quantitative Archival Reproducibility Digest `80e0724e98e68de288ef0e0feb5d77cddc905f7d1d302451900d38bdc953e12d`; Zenodo 15-file ZIP SHA-256 `9f508f0417588550a4ba367ffdfbd7f2816d891b5cb1aa7517c47dd764978562` (16-file MIT/DOI-stamped ZIP SHA-256 `73eb4637e76cdef9e682874c25c730e74f0687a8cdecb0005392d961c2d3c3eb`).

Appendix B. Claim-boundary checklist for readers and reviewers

Question	Evidence-bounded answer
Does the portfolio contain rigorous mathematical results?	Yes. It includes analytic theorems (LRSC rank(V) = p, discrete Loewner $12/(\pi^2 R)$ trace bound, exact Hopf & J_4 algebra), Lean 4 pointwise box/floor proofs (`BoundaryVerification.lean` & `ExistenceVerification.lean`), and finite computer-assisted interval/integer certificates under explicit assumptions.
What exactly do the Lean 4 modules prove in Programs 03 and 08?	They machine-check pointwise algebraic sign and linear-floor inequalities on the vector field at the 6 faces of $B(e)$ (9 theorems) and 8 faces of B_4 (17 theorems, `0 sorry`, `0 nlinarith`, verified in GitHub Actions CI). Continuous-time trajectory invariance $x(t) \in B_4$ and Grönwall positivity $\gamma(t) \geq \gamma(0)e^{-bt} > 0$ follow by classical Nagumo and ODE comparison theorems applied to those certified pointwise bounds.
What sustains the zero-drive ($P_0 = 0$) cycle in Program 08?	The slow self-modeling memory variable $\Psi(t)$ ($\mu_\Psi = 0.60$), which remains above $\Psi_{min} = 0.7305$ across the deep erasure trough so $\lambda_{lucid}\Psi \geq 0.219$ acts as a stored capacitive drive. The memoryless proportional loop alone ($\lambda_{sim} = 7.20$, $\lambda_{lucid} = 0$) decays to the locally stable origin (0,0,0,0).
Does the Navier–Stokes work solve the 3D continuum regularity problem?	No. The strongest statements are 128/192-bit Arb finite-cutoff ODE certificates (N = 11, …, 18 over T = 0.003 for one 112-pair rational datum). Continuum transfer and recursive backreaction drift remain open.
Does the Riemann WP84 work prove the Riemann Hypothesis or an unconditional > 79% bound?	No. The discrete Loewner trace bound, C_5 151-vertex polyhedral sign certificate, and 4th-cumulant counterexamples are finite/operator theorems. The 68.8203% (R = 400) and 79.6271% (11061/13891) figures are conditional/model constructions; the unconditional literature baseline remains > 67.25%, and RH remains open.
Does the AI work prove machine self-awareness, consciousness, or qualia?	No. Self-Referential Processing v8.6.3 is preregistered and paused; Phase 7 is synthetic GLMM calibration; Ψ v4 is a pre-confirmatory 2AFC diagnostic; Program 08's Ψ > 0.50 is a mathematical readout threshold on R_{self}.

Question	Evidence-bounded answer
Do the Physics or Bounded Existence ODEs prove a microscopic law of nature?	No. Both are dimensionless phenomenological ODE systems with no physical energy or entropy budget. Non-geometric Gate 0 (observer-to-horizon bridge) remains open.
Is LRSC $K_{0.001} = 11$ the same thing as the rank(V) = p theorem?	No. rank(V) = p is the analytic full-column rank theorem (42, 41, 40 at θ = 0.07, 0.08, 0.09 for M = 99), whereas $K_{0.001}$ = 14, 11, 12 is the minimum subset cardinality for relative ℓ_2 error ≤ 10^{-3} on those finite benchmarks.
Are failed prospective tests and negative results preserved?	Yes. The Navier N17 mechanism gate failure, BGT critical-space failure, Riemann $p_s \le F_{61}$ falsifier, and STRENGTH H4-BRIDGE-1 high-noise J75 failure (4/10 pass) are permanently retained in the record without retuning.
Can this dossier replace the primary repositories and frozen archives?	No. It is an evidence-bounded synthesis. The primary GitHub repositories, OSF/Zenodo deposits, SHA-256 manifests, and replay scripts remain authoritative for reproduction.

END OF MASTER RESEARCH DOSSIER

Prepared from the complete research continuity record, primary frozen OSF/Zenodo deposits, verified GitHub Actions CI runs, and the active 24 October 2026 portfolio state.